Hardware Design Specification (HDS)

Modular Chilled-Water Cooling Plant with Building Management System Integration

ISA/IEC 62443 Target Security Level 3 (SL-T 3) — hardware for FRS-CHW-001 and SDS-CHW-001

Template with worked example

Worked example: data center plant — Trane Symbio 800 chiller modules · Rockwell ControlLogix 5580 plant control · Siemens Desigo CC (primary) · Kepware historian (secondary, long-term)

Field	Value
Document number	**[HDS-CHW-001]**
Revision	**[A — issued for review]**
Parent documents	**[URS-CHW-001, FRS-CHW-001, SDS-CHW-001 Rev A]**
Facility	**[DC-01 data center, chilled-water plant]**
Asset owner	**[Owner organization]**
System integrator	**[Integrator organization]**
Prepared by	**[Name, role]**
Date	**[DD Month YYYY]**
Data classification	**[Business Confidential]** (example marking)

Vendor settings, product capabilities and catalog references are as of 28 October 2026; confirm them at FAT, SAT and handover.

Using this template

- Each hardware element names the FRS rows it supports; together with the SDS, Annex A shows every FRS row is designed.
- Catalog numbers are worked-example selections — confirm against current vendor data, sizing and lead times.
- Physical security hardware is part of the security design: it is the compensation for interfaces that cannot authenticate individuals.
- Black text is the template requirement. *Blue italic text is the worked example.* **[Bracketed blue text]** is a project fill-in with the worked-example default.
- This template is open source, with no copyright. The Business Confidential marking is an example of the owner's data classification; replace it with the owner's own label.

Approvals

Role	Name	Signature	Date
Asset owner — facilities / mechanical engineering	**[Name]**		**[Date]**
Asset owner — OT cybersecurity	**[Name]**		**[Date]**
System integrator — controls lead	**[Name]**		**[Date]**
System integrator — electrical lead	**[Name]**		**[Date]**

Revision history

Rev	Date	Description	Author
[A]	**[Date]**	**[Initial issue for review]**	**[Name]**

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1 Purpose and scope

This Hardware Design Specification (HDS) defines the physical design that carries the SDS: equipment list, control panels and chassis layout, I/O, network hardware and cabling, power, hosts, physical security, final-element fail positions and the hardware security features to enable. Catalog numbers are worked-example selections to be confirmed against current vendor data and project sizing.

Engineering drawings issued from this HDS shall have no blank or “unknown” fields; every field is filled or marked as a tracked open item.

2 Equipment list (worked example)

Tag	Item	Make / model	Qty	Zone	Type	Notes	FRS
CH-1…4	**Water-cooled chiller module**	*Trane chiller with Symbio 800 controller and TD7 display (factory-fitted)*	[4]	Z1-n	EDR	*No wireless or cellular options fitted.*	FRS-SEC-106, FRS-DEV-E213
CP-A, CP-B	**Plant control panel**	*See section 3*	[2]	Z2	—	*One per controller (DD-01).*	FRS-DEG-003
PLC-A, PLC-B	**Plant controller**	*ControlLogix 5580, 1756-L83E (confirm memory sizing)*	[2]	Z2	EDR	*Firmware per OI-03.*	FRS-DEG-003, FRS-INT-003
EN-A1, EN-B1	**CIP Security communication module**	*1756-EN4TR*	[2]	Z2	EDR	*Plant DLR ring and CIP Security.*	FRS-INT-010, FRS-NET-007
MB-A, MB-B	**Modbus TCP client module**	*ProSoft MVI56E-MNETC*	[2]	Z2	EDR	*Chiller polling (DD-03).*	FRS-INT-001
P-CHW-1…4	**CHW pump drives**	*PowerFlex 755T family (frame per motor kW)*	[4]	Z2	EDR	*CIP Security; HIM inside locked MCC section.*	FRS-HYD-001, FRS-INT-010
P-CW-1…4	**CW pump drives**	*PowerFlex 755T family*	[4]	Z2	EDR		FRS-HYD-005
F-CT-1…4	**Tower fan drives**	*Danfoss VLT HVAC Drive FC 102 with MCA 121 EtherNet/IP option*	[4]	Z2	EDR	*Vibration switch to safe-stop input.*	FRS-CT-001, FRS-CT-003
FT-CHW-1, 2	**CHW flowmeters**	*Endress+Hauser Proline Promag W 300 or 500 (EtherNet/IP)*	[2]	Z2	EDR	*Matched RTD pair per meter for thermal load.*	FRS-MET-002
M-CH-1…4	**Chiller feeder meters**	*Schneider PowerLogic PM8000 (PM8240)*	[4]	Z3	EDR		FRS-MET-001
M-MCC-1…4	**MCC feeder meters**	*Schneider PowerLogic PM5560*	[4]	Z3	EDR		FRS-MET-001
WT-1	**Water-treatment skid**	*Vendor skid: SIMATIC S7-1500 CPU (model per vendor), remote I/O, PROFIBUS DP dosing pumps, DP/PA coupler and PA instruments*	[1]	Z4	EDR	*Internal PROFINET / PROFIBUS not routed (OI-11).*	FRS-WT-001, FRS-WT-002
GW-1	**Refrigerant and leak panel gateway**	*ProSoft / FieldServer QuickServer PS-QS-3x10-F*	[1]	Z4	NDR		FRS-INT-007
RM-1	**Refrigerant monitor panel**	**[Vendor and model per OI-08]**	[1]	—	—	*Hardwired outputs to ventilation and alarms.*	FRS-SAF-001
PXC-1	**Plant-room HVAC automation station**	*Siemens Desigo PXC7*	[1]	Z10	EDR	*BACnet/SC; MS/TP trunk to room controllers.*	FRS-PRH-001, FRS-INT-008
OIS-1	**Local operator station**	*Cincoze CV-100 + P2102 panel PC with Windows IoT Enterprise LTSC, FactoryTalk View SE local station*	[1]	Z2	EDR + HDR + SAR	*Platform EDR, OS HDR, application SAR recorded separately.*	FRS-HMI-001
SW-R1…R6	**Plant ring switches**	*Rockwell Stratix 5800, 1783-MMS10 family (variant with DLR and PTP, confirm)*	[6]	Z1, Z2, Z3, Z4, Z10	NDR		FRS-NET-007, FRS-SEC-211
SW-C1, C2	**Supervisory and infrastructure switches**	*Cisco Catalyst 9300 (model per port count)*	[2]	Z5, Z6, Z7	NDR		FRS-NET-001
FW-P1, P2	**Plant boundary firewall HA pair**	*Fortinet FortiGate Rugged 70F (FortiOS 7.6)*	[2]	Boundary	NDR	*No bypass ports used.*	FRS-NET-002, FRS-NET-005
FW-D1, D2	**IDMZ firewall HA pair**	*Fortinet FortiGate 200F (FortiOS 7.6)*	[2]	Z8 boundary	NDR		FRS-SEC-113, FRS-NET-005
ESX-5A, 5B	**Z5 virtualization hosts**	**[2U server]**, *TPM 2.0, UEFI Secure Boot, ESXi 8*	[2]	Z5	HDR	*Desigo CC and Kepware VMs.*	FRS-SEC-504, FRS-DEV-H314

Tag	Item	Make / model	Qty	Zone	Type	Notes	FRS
ESX-7A, 7B	**Z7 virtualization hosts**	**[2U server]**, *TPM 2.0, UEFI Secure Boot, ESXi 8*	**[2]**	Z7	HDR	*Domain controllers, CA, collector, backup, monitoring console.*	FRS-SEC-504
EWS-1, 2	**Engineering workstations**	**[Workstation]** *with TPM 2.0, Windows 11*	**[2]**	Z6	HDR		FRS-LCM-004
NTP-1, 2	**GPS-disciplined time servers**	**[NTP appliance with GPS receiver]**	**[2]**	Z7	NDR	*Antenna route per section 6.*	FRS-SEC-211
JH-1	**Jump host**	**[Server or VM]** *with MFA and session recording*	**[1]**	Z8	HDR		FRS-SEC-113
MON-1…n	**Passive monitoring sensors**	**[Owner-selected platform]**	**[per SPAN]**	Z1–Z5, Z10	NDR	*SPAN or TAP, receive-only.*	FRS-SEC-602
UPS-C	**Controls UPS**	**[UPS sized for 15 min]**	**[2]**	—	—	*Generator-backed supply.*	FRS-DEG-005, FRS-SEC-705

3 Plant control panels CP-A and CP-B

Identical panels, one per controller (SDS DD-01). Each panel: 10-slot chassis, redundant power, door switch, keyed lock, terminal sections for field I/O and the hardwired permissive circuits, ring switch node and fibre patch.

Slot	Catalog	Function	Notes
0	*1756-L83E*	ControlLogix 5580 controller (A or B)	Keyswitch RUN; key held by Controls Engineer
1	*1756-EN4TR*	EtherNet/IP with CIP Security — DLR ring to Z2	Port to ring nodes; CIP Sync time
2	*MVI56E-MNETC*	Modbus TCP client — owned chillers and skid status	Client map per FRS-INT-001 and INT-006
3	*1756-IF8I*	Isolated analog inputs — DP, levels, valve positions	Signal conditioning per I/O list
4	*1756-IRT8I*	Isolated RTD inputs — CHW, CW, TES temperatures	Matched pairs for thermal load
5	*1756-OF8I*	Isolated analog outputs — valve positions	Fail position per section 9
6	*1756-IB16*	Digital inputs — run, flow, permissives, door, trips	Panel door switch and skid trip included
7	*1756-OB16E*	Digital outputs — enable permissives, TES discharge	Electronically fused
8–9	—	Spare	Kept free for expansion

Chassis 1756-A10 with redundant power (1756-PA75R pair with redundant power chassis adapter, confirm). Catalog numbers are worked-example selections.

4 I/O design and hardwired signals

Signals below are hardwired so that safety and critical status never depend on a network (FRS-INT-012). The full I/O list is a project deliverable.

Tag	Signal	Type	Connected to	Purpose	FRS
CH-n-EN	**Chiller n enable permissive**	DO → chiller enable circuit	PLC A/B	Hardwired enable in series with flow proving	FRS-CHL-002, FRS-SAF-006
CH-n-RUN	**Chiller n running**	DI	PLC A/B	Run proof independent of Modbus	FRS-CHL-002
FS-CHW-n / FS-CW-n	**Evaporator / condenser flow switch**	DI + series in enable circuit	PLC and chiller	Flow proving permissive	FRS-SAF-006, FRS-CHL-002

Tag	Signal	Type	Connected to	Purpose	FRS
ES-PLANT	**Emergency stop**	Safety relay; DI status	Hardwired to drives and chillers	Stops plant without any network	FRS-SAF-006
TES-DV	**TES discharge valve**	DO (spring return, fail open)	PLC A	Ride-through	FRS-TES-001, FRS-DEG-006
VIB-CT-n	**Tower fan vibration**	Wired to FC 102 safe-stop input	Drive	Trip without network	FRS-CT-003
RM-ALM	**Refrigerant alarm**	Volt-free contact → ventilation and horn; DI copy to PLC	Refrigerant monitor	Safety action hardwired; PLC monitors	FRS-SAF-001
FA-SD	**Fire alarm shutdown**	Relay to MCC and chiller stop circuits; DI copy to PLC	Fire alarm panel	Hardwired shutdown	FRS-SAF-003
WT-TRIP	**Skid summary trip**	DI	PLC A/B	Skid status independent of network	FRS-INT-006, FRS-INT-012
CP-n-DOOR	**Panel door switch**	DI	PLC A/B	Door-open alarm (physical access)	FRS-DEV-E213, FRS-DEV-E311
UPS-ALM	**UPS on battery / fault**	DI	PLC A/B	Power-event detection	FRS-DEG-005
PSU-ALM	**24 VDC supply redundancy lost**	DI	PLC A/B	Power monitoring	FRS-DEG-005

5 Final-element fail positions

Element	Actuator / hardware	Fail position (control power loss)	FRS
Chiller isolation valves	*Electric actuator, fail in place*	Hold	FRS-DEG-006
Minimum-flow bypass valve	*Electric actuator with spring return*	Open	FRS-DEG-006, FRS-HYD-002
Economizer HX valves	*Electric actuator with spring return*	Closed	FRS-DEG-006, FRS-ECO-002
TES discharge valve	*Electric actuator with spring return*	Open	FRS-DEG-006, FRS-TES-001
Tower bypass valve	*Electric actuator with spring return*	To tower	FRS-DEG-006, FRS-CT-004

6 Network hardware and cabling

ID	Element	Design (worked example)	FRS
NW-01	**Ring cabling**	*Z2 DLR ring on single-mode or OM4 multimode fibre between plant panels and module switches; each ring node in a locked panel.*	FRS-NET-007, FRS-DEG-004
NW-02	**Copper drops**	*Shielded Cat 6A from ring switches to drives, meters and chiller controllers; labelled at both ends with tag, zone and port.*	FRS-NET-001
NW-03	**Unused ports**	*Unused switch ports shut in a parking VLAN and fitted with keyed RJ45 port locks; unused fibre ports capped.*	FRS-SEC-707, FRS-DEV-N213
NW-04	**Boundary cabling**	*Each zone uses its own FortiGate interface; HA heartbeat links direct between units; no bypass modules.*	FRS-NET-005, FRS-NET-002
NW-05	**Monitoring taps**	*SPAN sessions on ring and core switches, or passive TAPs where SPAN load is a concern; sensors receive-only.*	FRS-SEC-602
NW-06	**Management network**	*Out-of-band management ports of switches, firewalls and server BMCs cabled only to Z7-M.*	FRS-NET-008, FRS-DEV-H213
NW-07	**Time distribution**	*GPS antennas with surge protection on roof; NTP appliances in Z7 rack; Stratix 5800 PTP boundary clock in Z2.*	FRS-SEC-211

7 Power design

ID	Element	Design (worked example)	FRS
PW-01	**Controls UPS**	*Two UPS units feeding A and B panels, network and servers separately; 15 min autonomy; generator-backed.*	FRS-DEG-005, FRS-SEC-705
PW-02	**Panel DC power**	*Redundant 24 VDC supplies with diode or redundancy module; loss of redundancy alarmed (PSU-ALM).*	FRS-DEG-005
PW-03	**Restart behaviour**	*All controllers, drives and switches return to configured state on power return; chiller restarts staggered by the plant program.*	FRS-DEG-005, FRS-SEC-705
PW-04	**Separation**	*A and B panels on different UPS and distribution so one power failure affects one controller only.*	FRS-DEG-003, FRS-INT-012

8 Hosts and racks

ID	Element	Design (worked example)	FRS
HS-01	**Platform security**	*TPM 2.0 enabled; UEFI Secure Boot on; BIOS/UEFI administrator password set; boot from internal disk only; unused USB disabled in firmware.*	FRS-DEV-H213, FRS-DEV-H312, FRS-DEV-H314, FRS-SEC-105
HS-02	**Storage**	*RAID 1 or 10; self-encrypting disks or BitLocker / VM encryption; Kepware historian volume encrypted.*	FRS-SEC-401, FRS-HMI-007
HS-03	**Out-of-band management**	*Server BMC on Z7-M only, local accounts changed, firmware current.*	FRS-NET-008, FRS-DEV-H213
HS-04	**Racks**	*OT racks in a locked OT room with badge access; rack doors locked; chassis intrusion detection enabled where supported.*	FRS-DEV-H311, FRS-DEV-N311

9 Physical security hardware

ID	Element	Design (worked example)	FRS
PH-01	**Panel locks**	*Plant panels keyed differently from electrical panels; keys issued by role.*	FRS-DEV-E311, FRS-SEC-203
PH-02	**Door switches**	*Door switch in every control and network panel wired to the controller (CP-n-DOOR); door-open alarm P3.*	FRS-DEV-E213, FRS-DEV-E311, FRS-DEV-N311
PH-03	**Tamper evidence**	*Numbered tamper-evident seals on meter covers, drive keypads, Symbio USB service-port covers; seal register kept.*	FRS-DEV-E311, FRS-DEV-E213
PH-04	**Port blockers**	*Keyed USB and RJ45 blockers on panel PCs, EWS and unused switch ports.*	FRS-SEC-203, FRS-DEV-E213, FRS-DEV-N213
PH-05	**Plant room access**	*Badge-controlled access with logging; compensates for panel-only interfaces (TD7, keypads).*	FRS-SEC-101, FRS-CMP-101
PH-06	**Wireless**	*No wireless hardware fitted in OT panels; chiller wireless options not ordered.*	FRS-SEC-106, FRS-DEV-N106, FRS-SEC-202

10 Hardware security features to enable

Device	Hardware feature	Required state	FRS
ControlLogix 5580	Keyswitch	RUN; key removed and held by Controls Engineer	FRS-PLC-002, FRS-SEC-304
ControlLogix 5580	Front USB and SD card	Disabled where firmware supports it; otherwise panel locked and door alarmed	FRS-DEV-E213, FRS-SEC-203

Device	Hardware feature	Required state	FRS
Symbio 800	USB service port	Behind locked panel door; sealed cover	FRS-DEV-E213
PowerFlex 755T / FC 102	HIM / LCP keypad	Inside locked MCC or panel; keypad power-up reset reachable only by Controls Engineer	FRS-DEV-E213, FRS-SAF-004
Stratix 5800	Console and SD slot	Inside locked panel; SD card presence checked at SAT	FRS-DEV-N213
FortiGate	Console and USB	Inside locked rack; USB auto-install disabled	FRS-DEV-N213
Servers and workstations	TPM, Secure Boot, firmware password	Enabled and set per HS-01	FRS-DEV-H314, FRS-DEV-H312
Panel PC (OIS-1)	UEFI Secure Boot, TPM	Enabled; BIOS review only in an approved reboot window	FRS-DEV-H314

11 Environment, labelling, spares and decommissioning

ID	Element	Design (worked example)	FRS
EN-01	**Enclosures**	*NEMA 12 / IP54 panels in the plant room; NEMA 4X / IP66 for outdoor tower panels; cooling sized for 40 °C ambient.*	FRS-DEG-004
EN-02	**Grounding and bonding**	*Single-point panel ground bar; shield grounding at one end; surge protection on outdoor and antenna circuits.*	FRS-NET-007
EN-03	**Labelling and drawings**	*Every device, cable and port labelled with tag, zone and conduit; drawings with no blank or unknown fields.*	FRS-SEC-708, FRS-NET-001
EN-04	**Spares**	*One spare of each controller, communication module, I/O module type, switch and drive control board, stored at the approved firmware baseline with hash record.*	FRS-LCM-002, FRS-SEC-704
EN-05	**Decommissioning**	*Controller hardware destroyed at end of life (no documented secure erase); host media sanitized.*	FRS-SEC-402, FRS-CMP-402

12 ISA/IEC 62443-4-2 requirements per hardware device (EDR, HDR, NDR)

Every embedded, host and network device in the equipment list shall meet each requirement below at SL-C 3, including the requirement enhancements specified for SL 3, by capability or by the named compensation (URS-C-002). Each table lists the devices it applies to; the per-product SL-3 checklist for each device is the verification procedure.

12.1 Embedded devices (EDR)

Tag	Device
CH-1…4	*Trane Symbio 800 (+TD7)*
PLC-A, PLC-B	*ControlLogix 5580*
EN-A1, EN-B1	*1756-EN4TR*
MB-A, MB-B	*MVI56E-MNETC*
P-CHW, P-CW	*PowerFlex 755T*
F-CT-1…4	*Danfoss FC 102*
FT-CHW-1, 2	*Proline Promag 300/500*
M-CH-1…4	*PowerLogic PM8000*
M-MCC-1…4	*PowerLogic PM5560*
WT-1	*SIMATIC S7-1500 skid*
PXC-1	*Desigo PXC7*
OIS-1	*Cincoze CV-100 + P2102 platform*

Requirements for these devices:

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
Component requirements (all component types)					
CR 1.1	Human user identification and authentication	All EDR devices	*Named local or directory-backed accounts on Symbio UI, ControlLogix (FactoryTalk Security), FC 102 SCF, PM8000, PM5560, S7-1500 user management, PXC7; OIS-1 BIOS admin.*	*TD7 display, drive keypads, meter displays: panel only → locked panels, door alarms, badge-controlled plant room (HDS PH-01, PH-02, PH-05).*	FRS-CMP-101
CR 1.2	Software process and device identification and authentication	All EDR devices	*CIP Security device identity on ControlLogix (EN4TR) and PowerFlex 755T; OPC UA certificates on ControlLogix and S7-1500; BACnet/SC on PXC7.*	*Symbio 800 (Modbus TCP, BACnet/IP), MVI56E-MNETC, PM8000, PM5560, FC 102 and Promag (if no CIP Security) → endpoint pinning at FortiGate (R-C1, R-C4).*	FRS-CMP-102
CR 1.3	Account management	All EDR devices	*Accounts managed per device; FactoryTalk Security and S7-1500 user management integrate with central policy where supported.*	*Device-local accounts (Symbio, FC 102, meters, PXC7) → account register, 90-day review.*	FRS-CMP-103
CR 1.4	Identifier management	All EDR devices	*Unique identifiers per person; service identities named by role.*	—	FRS-CMP-104
CR 1.5	Authenticator management	All EDR devices	*All factory credentials changed before SAT; certificate private keys in hardware where supported.*	*No hardware key store on several EDRs → recorded gap; conduit restriction.*	FRS-CMP-105
CR 1.7	Strength of password-based authentication	All EDR devices	*Password strength set where configurable (S7-1500, PM5560 4.x, FC 102 SCF).*	*Numeric display passcodes and keypad codes → physical control.*	FRS-CMP-107

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
CR 1.8	Public key infrastructure certificates	All EDR devices	*Owner-PKI certificates on HTTPS, OPC UA, CIP Security and BACnet/SC endpoints.*	*Devices that cannot import owner certificates → written risk acceptance.*	FRS-CMP-108
CR 1.9	Strength of public key-based authentication	All EDR devices	*Chain and revocation checks in OPC UA and CIP Security stacks.*	*Hardware key protection absent on most EDRs → recorded gap.*	FRS-CMP-109
CR 1.10	Authenticator feedback	All EDR devices	*Masked entry on device web UIs and tools.*	—	FRS-CMP-110
CR 1.11	Unsuccessful login attempts	All EDR devices	*Lockout or delay where supported (FC 102 0-94, S7-1500, PM5560 4.x).*	*Devices without lockout → FortiGate rate limiting and monitoring.*	FRS-CMP-111
CR 1.12	System use notification	All EDR devices	*Banner where supported (PM5560 4.x).*	*Most EDRs have no banner → recorded.*	FRS-CMP-112
CR 1.14	Strength of symmetric key-based authentication	All EDR devices	*No shared symmetric keys stored on EDRs except NTP keys where supported.*	—	FRS-CMP-114
CR 2.1	Authorization enforcement	All EDR devices	*Role permissions: Symbio UI, FactoryTalk Security, FC 102 SCF roles, S7-1500 roles; keyswitch RUN on ControlLogix.*	*Single-role devices (meters, Promag) → Z6-only engineering access and physical control.*	FRS-CMP-201
CR 2.2	Wireless use control	All EDR devices	*No wireless fitted (Symbio Air-Fi, Wi-Fi, cellular not ordered).*	—	FRS-CMP-202
CR 2.5	Session lock	All EDR devices	*Web UI session timeouts where supported.*	—	FRS-CMP-205
CR 2.6	Remote session termination	All EDR devices	*Remote session timeouts where supported; engineering sessions only from Z6.*	—	FRS-CMP-206
CR 2.7	Concurrent session control	All EDR devices	*Session limits where supported; controller OPC UA session count controlled (FRS-INT-004).*	—	FRS-CMP-207
CR 2.8	Auditable events	All EDR devices	*Symbio audit trail; ControlLogix controller log and change detection; FC 102 audit log 43-6*; S7-1500 diagnostic events; PM8000 event log.*	*MVI56E-MNETC, Promag and PM5560 limited logs → boundary and network-monitoring logs.*	FRS-CMP-208
CR 2.9	Audit storage capacity	All EDR devices	*Device log depth recorded; logs collected off-device.*	*Small device buffers → collection frequency set to avoid overwrite.*	FRS-CMP-209
CR 2.10	Response to audit processing failures	All EDR devices	*Device continues control if logging fails (inherent).*	—	FRS-CMP-210
CR 2.11	Timestamps	All EDR devices	*NTP on Symbio, meters, S7-1500, PXC7; CIP Sync (PTP) on ControlLogix.*	*Devices without time sync → collector receive time.*	FRS-CMP-211
CR 2.12	Non-repudiation	All EDR devices	*Named logins plus MOC comments in Studio 5000 and TIA projects.*	—	FRS-CMP-212
CR 3.1	Communication integrity	All EDR devices	*CIP Security, OPC UA SignAndEncrypt, BACnet/SC, HTTPS.*	*Modbus TCP, BACnet/IP, EtherNet/IP without CIP Security, PROFINET, PROFIBUS → confined per URS-Z-005.*	FRS-CMP-301
CR 3.3	Security functionality verification	All EDR devices	*Configuration exports compared with baselines.*	—	FRS-CMP-303
CR 3.4	Software and information integrity	All EDR devices	*ControlLogix change detection; FC 102 0-91 / 0-92; signed firmware (Symbio, ControlLogix, 755T, S7-1500).*	*Devices without integrity checks → periodic configuration comparison.*	FRS-CMP-304
CR 3.5	Input validation	All EDR devices	*Controller-side validation (FRS-PLC-008); device range limits.*	—	FRS-CMP-305
CR 3.6	Deterministic output	All EDR devices	*Fail states per FRS-DEG table and drive communication-loss actions.*	—	FRS-CMP-306
CR 3.7	Error handling	All EDR devices	*Generic web and protocol error responses.*	—	FRS-CMP-307
CR 3.8	Session integrity	All EDR devices	*Native session handling on device web servers.*	—	FRS-CMP-308
CR 3.9	Protection of audit information	All EDR devices	*Device logs forwarded off-device.*	—	FRS-CMP-309

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
CR 4.1	Information confidentiality	All EDR devices	*Encrypted Symbio backups; project files on encrypted storage.*	*Clear-text configuration exports (e.g., PM5560 Config.csv) → handling rules.*	FRS-CMP-401
CR 4.2	Information persistence	All EDR devices	*Factory reset plus destruction at end of life.*	*ControlLogix: no documented secure erase → destroy.*	FRS-CMP-402
CR 4.3	Use of cryptography	All EDR devices	*TLS 1.2+ and current OPC UA policies on EDR endpoints.*	—	FRS-CMP-403
CR 5.1	Network segmentation	All EDR devices	*Each EDR on its zone network only.*	—	FRS-CMP-501
CR 6.1	Audit log accessibility	All EDR devices	*Logs readable via tools or forwarding.*	—	FRS-CMP-601
CR 6.2	Continuous monitoring	All EDR devices	*Passive monitoring covers all EDR networks.*	—	FRS-CMP-602
CR 7.1	Denial of service protection	All EDR devices	*Storm control on the switch ports serving EDRs; chiller local autonomy.*	—	FRS-CMP-701
CR 7.2	Resource management	All EDR devices	*Controller resources trended (FRS-PLC-016 to 019).*	—	FRS-CMP-702
CR 7.3	Control system backup	All EDR devices	*Backups: .ACD/.L5X, Symbio automated backup, MCT 10, TIA project, DeviceCare, ION Setup, ABT.*	—	FRS-CMP-703
CR 7.4	Control system recovery and reconstitution	All EDR devices	*Spares at baseline; restore runbooks.*	—	FRS-CMP-704
CR 7.6	Network and security configuration settings	All EDR devices	*Baseline per product checklist; exported configurations.*	—	FRS-CMP-706
CR 7.7	Least functionality	All EDR devices	*Unused ports, protocols and services off (ControlLogix web, FC 102 SNMP, Symbio LON, S7 PUT/GET).*	*Services that cannot be disabled → blocked at the boundary.*	FRS-CMP-707
CR 7.8	Control system component inventory	All EDR devices	*Model, catalog and firmware recorded per device.*	—	FRS-CMP-708
EDR-specific requirements					
EDR 2.4	Mobile code	All EDR devices	*Symbio UI and PowerLogic web pages used only from Z6 hosts; ControlLogix web pages off.*	—	FRS-DEV-E204
EDR 2.13	Use of physical diagnostic and test interfaces	All EDR devices	*Symbio USB service port, drive keypads and USB, ControlLogix USB and SD, meter ports behind locked doors with door-open alarms to the plant controller.*	*Ports that cannot be disabled → physical control and alarm.*	FRS-DEV-E213
EDR 3.2	Protection from malicious code	All EDR devices	*Only signed firmware accepted (Symbio 800, ControlLogix, PowerFlex 755T, S7-1500).*	*Devices without signature checks → firmware hash verified before load (URS-L-010).*	FRS-DEV-E302
EDR 3.10	Support for updates	All EDR devices	*Firmware updated module by module under MOC with hash verification.*	—	FRS-DEV-E310
EDR 3.11	Physical tamper resistance and detection	All EDR devices	*Locked enclosures with door switches; tamper-evident seals on meters.*	—	FRS-DEV-E311
EDR 3.12	Provisioning product supplier roots of trust	All EDR devices	*Vendor firmware-signing roots of trust per product (confirm, Annex D).*	—	FRS-DEV-E312
EDR 3.13	Provisioning asset owner roots of trust	All EDR devices	*Owner CA loaded for HTTPS, OPC UA and CIP Security where supported.*	*Devices that cannot load owner CA → risk acceptance.*	FRS-DEV-E313
EDR 3.14	Integrity of the boot process	All EDR devices	*Boot-time firmware verification where supported.*	*Gaps recorded per product.*	FRS-DEV-E314

Applies to: CH-1…4; PLC-A, PLC-B; EN-A1, EN-B1; MB-A, MB-B; P-CHW, P-CW; F-CT-1…4; FT-CHW-1, 2; M-CH-1…4; M-MCC-1…4; WT-1; PXC-1; OIS-1.

12.2 Host devices (HDR)

Tag	Device
ESX-5A/5B, ESX-7A/7B	*VMware ESXi 8 hosts*
Z5/Z7/Z8 servers	*Windows Server 2026 (Desigo CC, Kepware, DCs, CA, collector, backup, JH-1)*
EWS-1, 2; operator clients	*Windows 11*
OIS-1	*Windows IoT Enterprise LTSC on panel PC*

Requirements for these devices:

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
Component requirements (all component types)					
CR 1.1	Human user identification and authentication	All HDR devices	*OT domain accounts; ESXi via domain or vCenter SSO; console logon by named users.*	—	FRS-CMP-101
CR 1.2	Software process and device identification and authentication	All HDR devices	*Machine certificates; Kerberos for domain hosts.*	—	FRS-CMP-102
CR 1.3	Account management	All HDR devices	*OT domain account management; Windows LAPS for local admins.*	*ESXi local root → vaulted, break-glass only.*	FRS-CMP-103
CR 1.4	Identifier management	All HDR devices	*Unique identifiers.*	—	FRS-CMP-104
CR 1.5	Authenticator management	All HDR devices	*Credentials protected by TPM 2.0; Credential Guard.*	—	FRS-CMP-105
CR 1.7	Strength of password-based authentication	All HDR devices	*Domain password policy; ESXi Security.PasswordQualityControl.*	—	FRS-CMP-107
CR 1.8	Public key infrastructure certificates	All HDR devices	*Owner-PKI machine and TLS certificates.*	—	FRS-CMP-108
CR 1.9	Strength of public key-based authentication	All HDR devices	*Windows and ESXi certificate validation with OT CRL.*	—	FRS-CMP-109
CR 1.10	Authenticator feedback	All HDR devices	*Masked entry.*	—	FRS-CMP-110
CR 1.11	Unsuccessful login attempts	All HDR devices	*Domain lockout policy; ESXi Security.AccountLockFailures and AccountUnlockTime.*	—	FRS-CMP-111
CR 1.12	System use notification	All HDR devices	*Legal notice GPO; ESXi login message.*	—	FRS-CMP-112
CR 1.14	Strength of symmetric key-based authentication	All HDR devices	*No shared symmetric keys outside the Z7 key store.*	—	FRS-CMP-114
CR 2.1	Authorization enforcement	All HDR devices	*Role groups per SDS 10; least-privilege service accounts.*	—	FRS-CMP-201
CR 2.2	Wireless use control	All HDR devices	*Wireless adapters absent or disabled in firmware.*	—	FRS-CMP-202
CR 2.5	Session lock	All HDR devices	*15-min screen lock; operator consoles view-only lock.*	—	FRS-CMP-205
CR 2.6	Remote session termination	All HDR devices	*RDP and shell idle timeouts; ESXi UserVars.ESXiShellInteractiveTimeOut.*	—	FRS-CMP-206
CR 2.7	Concurrent session control	All HDR devices	*RDP session limits per host.*	—	FRS-CMP-207
CR 2.8	Auditable events	All HDR devices	*Advanced audit policy; event forwarding; ESXi Syslog.global.logHost.*	—	FRS-CMP-208
CR 2.9	Audit storage capacity	All HDR devices	*Local log size set; collector capacity.*	—	FRS-CMP-209

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
CR 2.10	Response to audit processing failures	All HDR devices	*Forwarding failure alarmed.*	—	FRS-CMP-210
CR 2.11	Timestamps	All HDR devices	*Domain time from Z7 NTP; ESXi NTP.*	—	FRS-CMP-211
CR 2.12	Non-repudiation	All HDR devices	*Named logins and change records.*	—	FRS-CMP-212
CR 3.1	Communication integrity	All HDR devices	*SMB signing, RDP with TLS and NLA, LDAPS, HTTPS.*	—	FRS-CMP-301
CR 3.3	Security functionality verification	All HDR devices	*Configuration compliance scans against baseline.*	—	FRS-CMP-303
CR 3.4	Software and information integrity	All HDR devices	*File-integrity monitoring; application control; ESXi Secure Boot and attestation.*	—	FRS-CMP-304
CR 3.5	Input validation	All HDR devices	*Applications validate inputs (SAR responsibility).*	—	FRS-CMP-305
CR 3.6	Deterministic output	All HDR devices	*Hosts fail without affecting plant control (control stays in Z1–Z4).*	—	FRS-CMP-306
CR 3.7	Error handling	All HDR devices	*Generic error pages on hosted services.*	—	FRS-CMP-307
CR 3.8	Session integrity	All HDR devices	*Native session handling.*	—	FRS-CMP-308
CR 3.9	Protection of audit information	All HDR devices	*Logs forwarded; local log access restricted.*	—	FRS-CMP-309
CR 4.1	Information confidentiality	All HDR devices	*BitLocker or VM encryption; encrypted historian volume.*	—	FRS-CMP-401
CR 4.2	Information persistence	All HDR devices	*Media sanitization at decommissioning.*	—	FRS-CMP-402
CR 4.3	Use of cryptography	All HDR devices	*TLS 1.2+; legacy protocols (SMBv1, NTLMv1) disabled.*	—	FRS-CMP-403
CR 5.1	Network segmentation	All HDR devices	*Hosts placed by zone; separate ESXi host pairs for Z5 and Z7.*	—	FRS-CMP-501
CR 6.1	Audit log accessibility	All HDR devices	*Collector API access.*	—	FRS-CMP-601
CR 6.2	Continuous monitoring	All HDR devices	*Endpoint and network monitoring.*	—	FRS-CMP-602
CR 7.1	Denial of service protection	All HDR devices	*Host firewall; service hardening.*	—	FRS-CMP-701
CR 7.2	Resource management	All HDR devices	*Resource monitoring; endpoint-protection exclusions per vendor.*	—	FRS-CMP-702
CR 7.3	Control system backup	All HDR devices	*Image and configuration backups (SVC-06).*	—	FRS-CMP-703
CR 7.4	Control system recovery and reconstitution	All HDR devices	*Gold images; restore runbooks.*	—	FRS-CMP-704
CR 7.6	Network and security configuration settings	All HDR devices	*CIS-style baselines from per-product checklists; exported settings.*	—	FRS-CMP-706
CR 7.7	Least functionality	All HDR devices	*Unused roles, services and ports removed; ESXi SSH and shell off.*	—	FRS-CMP-707
CR 7.8	Control system component inventory	All HDR devices	*Inventory with OS build and patch level.*	—	FRS-CMP-708
HDR-specific requirements					
HDR 2.4	Mobile code	All HDR devices	*Application allowlisting; signed scripts only.*	—	FRS-DEV-H204
HDR 2.13	Use of physical diagnostic and test interfaces	All HDR devices	*UEFI password; unused USB disabled; out-of-band management only in Z7.*	—	FRS-DEV-H213
HDR 3.2	Protection from malicious code	All HDR devices	*Vendor-qualified endpoint protection or allowlisting.*	—	FRS-DEV-H302
HDR 3.10	Support for updates	All HDR devices	*Patches staged in Z8 after vendor qualification; applied host by host.*	—	FRS-DEV-H310

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
HDR 3.11	Physical tamper resistance and detection	All HDR devices	*Locked racks; chassis intrusion alarm where supported.*	—	FRS-DEV-H311
HDR 3.12	Provisioning product supplier roots of trust	All HDR devices	*UEFI Secure Boot with vendor keys.*	—	FRS-DEV-H312
HDR 3.13	Provisioning asset owner roots of trust	All HDR devices	*Owner certificates in machine stores.*	—	FRS-DEV-H313
HDR 3.14	Integrity of the boot process	All HDR devices	*UEFI Secure Boot and TPM measured boot; ESXi attestation.*	—	FRS-DEV-H314

Applies to: ESX-5A/5B, ESX-7A/7B; Z5/Z7/Z8 servers; EWS-1, 2; operator clients; OIS-1.

12.3 Network devices (NDR)

Tag	Device
SW-R1…R6	*Stratix 5800*
SW-C1, C2	*Catalyst 9300*
FW-P1, P2	*FortiGate Rugged 70F*
FW-D1, D2	*FortiGate 200F*
GW-1	*QuickServer PS-QS-3x10-F*
NTP-1, 2	*GPS NTP appliances*
MON-1…n	*Passive monitoring sensors*

Requirements for these devices:

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
Component requirements (all component types)					
CR 1.1	Human user identification and authentication	All NDR devices	*Named admins via TACACS+ / RADIUS (Stratix, Catalyst, FortiGate); FS-GUI users (QuickServer); appliance admins (NTP, sensors).*	—	FRS-CMP-101
CR 1.2	Software process and device identification and authentication	All NDR devices	*802.1X where used; certificate-based management TLS.*	—	FRS-CMP-102
CR 1.3	Account management	All NDR devices	*Central AAA; local fallback account vaulted.*	*QuickServer, NTP and sensor local accounts → register.*	FRS-CMP-103
CR 1.4	Identifier management	All NDR devices	*Unique identifiers.*	—	FRS-CMP-104
CR 1.5	Authenticator management	All NDR devices	*Bootstrap and label passwords changed (QuickServer label password).*	—	FRS-CMP-105
CR 1.7	Strength of password-based authentication	All NDR devices	*security passwords min-length; aaa common-criteria policy; FortiGate password policy.*	—	FRS-CMP-107
CR 1.8	Public key infrastructure certificates	All NDR devices	*Owner-PKI certificates for management HTTPS and BACnet/SC where relevant.*	—	FRS-CMP-108
CR 1.9	Strength of public key-based authentication	All NDR devices	*Certificate validation on management and VPN functions.*	—	FRS-CMP-109

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
CR 1.10	Authenticator feedback	All NDR devices	*Masked entry.*	—	FRS-CMP-110
CR 1.11	Unsuccessful login attempts	All NDR devices	*login block-for; FortiGate admin-lockout; QuickServer native lockout.*	—	FRS-CMP-111
CR 1.12	System use notification	All NDR devices	*banner login; FortiGate pre-login banner.*	—	FRS-CMP-112
CR 1.14	Strength of symmetric key-based authentication	All NDR devices	*NTP authentication keys; RADIUS/TACACS+ secrets per device pair, stored in Z7 vault.*	—	FRS-CMP-114
CR 2.1	Authorization enforcement	All NDR devices	*Privilege levels and admin profiles per role.*	—	FRS-CMP-201
CR 2.2	Wireless use control	All NDR devices	*No wireless on network devices.*	—	FRS-CMP-202
CR 2.5	Session lock	All NDR devices	*exec-timeout; FortiGate admintimeout.*	—	FRS-CMP-205
CR 2.6	Remote session termination	All NDR devices	*Idle and on-demand termination.*	—	FRS-CMP-206
CR 2.7	Concurrent session control	All NDR devices	*Session limits per admin.*	—	FRS-CMP-207
CR 2.8	Auditable events	All NDR devices	*Syslog (TLS where supported); FortiGate traffic and event logs.*	*QuickServer: no audit log documented → boundary logs and monitoring.*	FRS-CMP-208
CR 2.9	Audit storage capacity	All NDR devices	*Local buffers and forwarding.*	—	FRS-CMP-209
CR 2.10	Response to audit processing failures	All NDR devices	*Forwarding failure alarmed.*	—	FRS-CMP-210
CR 2.11	Timestamps	All NDR devices	*NTP with authentication; PTP on Stratix 5800.*	—	FRS-CMP-211
CR 2.12	Non-repudiation	All NDR devices	*Named admin accounts and change comments.*	—	FRS-CMP-212
CR 3.1	Communication integrity	All NDR devices	*SSH, HTTPS, SNMPv3; BACnet/SC through FortiGate untouched (end-to-end).*	*QuickServer BACnet/IP and Modbus RTU → confined.*	FRS-CMP-301
CR 3.3	Security functionality verification	All NDR devices	*Configuration compliance checks.*	—	FRS-CMP-303
CR 3.4	Software and information integrity	All NDR devices	*Image authenticity checks (IOS-XE, FortiOS); configuration change logging (archive log config).*	—	FRS-CMP-304
CR 3.5	Input validation	All NDR devices	*Protocol-aware filtering where supported (OI-14).*	—	FRS-CMP-305
CR 3.6	Deterministic output	All NDR devices	*FortiGate fail-close; switch ports default to defined VLAN.*	—	FRS-CMP-306
CR 3.7	Error handling	All NDR devices	*Generic error responses.*	—	FRS-CMP-307
CR 3.8	Session integrity	All NDR devices	*Native session handling.*	—	FRS-CMP-308
CR 3.9	Protection of audit information	All NDR devices	*Logs forwarded; local log access restricted.*	—	FRS-CMP-309
CR 4.1	Information confidentiality	All NDR devices	*Encrypted configuration backups (FortiGate); type 8/9 password storage on IOS-XE.*	—	FRS-CMP-401
CR 4.2	Information persistence	All NDR devices	*write erase / factory reset plus sanitization at end of life.*	—	FRS-CMP-402
CR 4.3	Use of cryptography	All NDR devices	*Strong crypto only; legacy SSH ciphers and SNMPv1/v2c off.*	—	FRS-CMP-403
CR 5.1	Network segmentation	All NDR devices	*VLANs and FortiGate interfaces per zone; Z2 DLR ring.*	—	FRS-CMP-501
CR 6.1	Audit log accessibility	All NDR devices	*Syslog and API access to logs.*	—	FRS-CMP-601
CR 6.2	Continuous monitoring	All NDR devices	*SPAN or TAP feeds to monitoring sensors.*	—	FRS-CMP-602
CR 7.1	Denial of service protection	All NDR devices	*Storm control, control-plane protection, FortiGate DoS policies.*	—	FRS-CMP-701

Ref.	Requirement	Applies to	Design realization (worked example)	Device-specific gaps and compensation	FRS
CR 7.2	Resource management	All NDR devices	*CPU and memory thresholds alarmed.*	—	FRS-CMP-702
CR 7.3	Control system backup	All NDR devices	*Automated configuration backups.*	—	FRS-CMP-703
CR 7.4	Control system recovery and reconstitution	All NDR devices	*Spare units at baseline; restore runbooks.*	—	FRS-CMP-704
CR 7.6	Network and security configuration settings	All NDR devices	*running-config and .conf exports compared with baseline.*	—	FRS-CMP-706
CR 7.7	Least functionality	All NDR devices	*HTTP server, Telnet and unused services off; unused ports shut.*	—	FRS-CMP-707
CR 7.8	Control system component inventory	All NDR devices	*show version / inventory exports into the inventory.*	—	FRS-CMP-708
NDR-specific requirements					
NDR 1.6	Wireless access management	All NDR devices	*No wireless on switches or firewalls.*	—	FRS-DEV-N106
NDR 1.13	Access via untrusted networks	All NDR devices	*FortiGate IDMZ pair with per-session approval through the jump host.*	—	FRS-DEV-N113
NDR 2.4	Mobile code	All NDR devices	*Management web UIs only from Z7.*	—	FRS-DEV-N204
NDR 2.13	Use of physical diagnostic and test interfaces	All NDR devices	*Console and USB ports inside locked cabinets.*	—	FRS-DEV-N213
NDR 3.2	Protection from malicious code	All NDR devices	*Signed IOS-XE and FortiOS images only.*	—	FRS-DEV-N302
NDR 3.10	Support for updates	All NDR devices	*Verified image upgrades, one unit of a redundant pair at a time.*	—	FRS-DEV-N310
NDR 3.11	Physical tamper resistance and detection	All NDR devices	*Locked network cabinets with door alarms.*	—	FRS-DEV-N311
NDR 3.12	Provisioning product supplier roots of trust	All NDR devices	*Vendor image-signing roots of trust.*	—	FRS-DEV-N312
NDR 3.13	Provisioning asset owner roots of trust	All NDR devices	*Owner CA for management HTTPS and RADIUS/TLS.*	—	FRS-DEV-N313
NDR 3.14	Integrity of the boot process	All NDR devices	*Image validation at boot.*	—	FRS-DEV-N314
NDR 5.2	Zone boundary protection	All NDR devices	*FortiGate deny-by-default, island mode and fail-close (FRS-NET-002, NET-004, NET-005).*	—	FRS-DEV-N502
NDR 5.3	General purpose person-to-person communication restrictions	All NDR devices	*FortiGate application control blocks person-to-person protocols.*	—	FRS-DEV-N503

Applies to: SW-R1…R6; SW-C1, C2; FW-P1, P2; FW-D1, D2; GW-1; NTP-1, 2; MON-1…n.

Annex A FRS → HDS coverage (with SDS)

Generated from the FRS, SDS and HDS data. FRS rows with hardware content are listed with their HDS elements (126 rows). FRS rows designed in neither the SDS nor the HDS: none. The full row-by-row view of both documents is SDS Annex A.

FRS row	HDS elements
FRS-CHL-002	CH-n-EN, CH-n-RUN, FS-CHW-n / FS-CW-n
FRS-HYD-001	P-CHW-1…4
FRS-HYD-002	HDS-5 Minimum-flow bypass valve
FRS-HYD-005	P-CW-1…4
FRS-CT-001	F-CT-1…4
FRS-CT-003	F-CT-1…4, VIB-CT-n
FRS-CT-004	HDS-5 Tower bypass valve
FRS-ECO-002	HDS-5 Economizer HX valves
FRS-TES-001	TES-DV, HDS-5 TES discharge valve
FRS-WT-001	WT-1
FRS-WT-002	WT-1
FRS-PRH-001	PXC-1
FRS-MET-001	M-CH-1…4, M-MCC-1…4
FRS-MET-002	FT-CHW-1, 2
FRS-DEG-003	CP-A, CP-B, PLC-A, PLC-B, HDS-3, PW-04
FRS-DEG-004	NW-01, EN-01
FRS-DEG-005	UPS-C, UPS-ALM, PSU-ALM, PW-01, PW-02, PW-03
FRS-DEG-006	TES-DV, HDS-5 Chiller isolation valves, HDS-5 Minimum-flow bypass valve, HDS-5 Economizer HX valves, HDS-5 TES discharge valve, HDS-5 Tower bypass valve
FRS-SAF-001	RM-1, RM-ALM
FRS-SAF-003	FA-SD
FRS-SAF-004	HDS-10 PowerFlex 755T / FC 102
FRS-SAF-006	CH-n-EN, FS-CHW-n / FS-CW-n, ES-PLANT
FRS-LCM-002	EN-04
FRS-LCM-004	EWS-1, 2
FRS-HMI-001	OIS-1, HDS-3
FRS-HMI-007	HS-02
FRS-INT-001	MB-A, MB-B, HDS-3
FRS-INT-003	PLC-A, PLC-B
FRS-INT-006	WT-TRIP
FRS-INT-007	GW-1
FRS-INT-008	PXC-1
FRS-INT-010	EN-A1, EN-B1, P-CHW-1…4, HDS-3
FRS-INT-012	WT-TRIP, HDS-4, PW-04
FRS-NET-001	SW-C1, C2, NW-02, EN-03
FRS-NET-002	FW-P1, P2, NW-04
FRS-NET-005	FW-P1, P2, FW-D1, D2, NW-04
FRS-NET-007	EN-A1, EN-B1, SW-R1…R6, NW-01, EN-02
FRS-NET-008	NW-06, HS-03
FRS-SEC-101	PH-05
FRS-SEC-105	HS-01
FRS-SEC-106	CH-1…4, PH-06

FRS row	HDS elements
FRS-SEC-113	FW-D1, D2, JH-1
FRS-SEC-202	PH-06
FRS-SEC-203	PH-01, PH-04, HDS-10 ControlLogix 5580
FRS-SEC-211	SW-R1…R6, NTP-1, 2, NW-07
FRS-SEC-304	HDS-10 ControlLogix 5580
FRS-SEC-401	HS-02
FRS-SEC-402	EN-05
FRS-SEC-504	ESX-5A, 5B, ESX-7A, 7B
FRS-SEC-602	MON-1…n, NW-05
FRS-SEC-704	EN-04
FRS-SEC-705	UPS-C, PW-01, PW-03
FRS-SEC-707	NW-03
FRS-SEC-708	EN-03
FRS-CMP-101	PH-05, HDS-12 EDR CR 1.1, HDS-12 HDR CR 1.1, HDS-12 NDR CR 1.1
FRS-CMP-102	HDS-12 EDR CR 1.2, HDS-12 HDR CR 1.2, HDS-12 NDR CR 1.2
FRS-CMP-103	HDS-12 EDR CR 1.3, HDS-12 HDR CR 1.3, HDS-12 NDR CR 1.3
FRS-CMP-104	HDS-12 EDR CR 1.4, HDS-12 HDR CR 1.4, HDS-12 NDR CR 1.4
FRS-CMP-105	HDS-12 EDR CR 1.5, HDS-12 HDR CR 1.5, HDS-12 NDR CR 1.5
FRS-CMP-107	HDS-12 EDR CR 1.7, HDS-12 HDR CR 1.7, HDS-12 NDR CR 1.7
FRS-CMP-108	HDS-12 EDR CR 1.8, HDS-12 HDR CR 1.8, HDS-12 NDR CR 1.8
FRS-CMP-109	HDS-12 EDR CR 1.9, HDS-12 HDR CR 1.9, HDS-12 NDR CR 1.9
FRS-CMP-110	HDS-12 EDR CR 1.10, HDS-12 HDR CR 1.10, HDS-12 NDR CR 1.10
FRS-CMP-111	HDS-12 EDR CR 1.11, HDS-12 HDR CR 1.11, HDS-12 NDR CR 1.11
FRS-CMP-112	HDS-12 EDR CR 1.12, HDS-12 HDR CR 1.12, HDS-12 NDR CR 1.12
FRS-CMP-114	HDS-12 EDR CR 1.14, HDS-12 HDR CR 1.14, HDS-12 NDR CR 1.14
FRS-CMP-201	HDS-12 EDR CR 2.1, HDS-12 HDR CR 2.1, HDS-12 NDR CR 2.1
FRS-CMP-202	HDS-12 EDR CR 2.2, HDS-12 HDR CR 2.2, HDS-12 NDR CR 2.2
FRS-CMP-205	HDS-12 EDR CR 2.5, HDS-12 HDR CR 2.5, HDS-12 NDR CR 2.5
FRS-CMP-206	HDS-12 EDR CR 2.6, HDS-12 HDR CR 2.6, HDS-12 NDR CR 2.6
FRS-CMP-207	HDS-12 EDR CR 2.7, HDS-12 HDR CR 2.7, HDS-12 NDR CR 2.7
FRS-CMP-208	HDS-12 EDR CR 2.8, HDS-12 HDR CR 2.8, HDS-12 NDR CR 2.8
FRS-CMP-209	HDS-12 EDR CR 2.9, HDS-12 HDR CR 2.9, HDS-12 NDR CR 2.9
FRS-CMP-210	HDS-12 EDR CR 2.10, HDS-12 HDR CR 2.10, HDS-12 NDR CR 2.10
FRS-CMP-211	HDS-12 EDR CR 2.11, HDS-12 HDR CR 2.11, HDS-12 NDR CR 2.11
FRS-CMP-212	HDS-12 EDR CR 2.12, HDS-12 HDR CR 2.12, HDS-12 NDR CR 2.12
FRS-CMP-301	HDS-12 EDR CR 3.1, HDS-12 HDR CR 3.1, HDS-12 NDR CR 3.1
FRS-CMP-303	HDS-12 EDR CR 3.3, HDS-12 HDR CR 3.3, HDS-12 NDR CR 3.3
FRS-CMP-304	HDS-12 EDR CR 3.4, HDS-12 HDR CR 3.4, HDS-12 NDR CR 3.4
FRS-CMP-305	HDS-12 EDR CR 3.5, HDS-12 HDR CR 3.5, HDS-12 NDR CR 3.5
FRS-CMP-306	HDS-12 EDR CR 3.6, HDS-12 HDR CR 3.6, HDS-12 NDR CR 3.6
FRS-CMP-307	HDS-12 EDR CR 3.7, HDS-12 HDR CR 3.7, HDS-12 NDR CR 3.7
FRS-CMP-308	HDS-12 EDR CR 3.8, HDS-12 HDR CR 3.8, HDS-12 NDR CR 3.8
FRS-CMP-309	HDS-12 EDR CR 3.9, HDS-12 HDR CR 3.9, HDS-12 NDR CR 3.9
FRS-CMP-401	HDS-12 EDR CR 4.1, HDS-12 HDR CR 4.1, HDS-12 NDR CR 4.1
FRS-CMP-402	EN-05, HDS-12 EDR CR 4.2, HDS-12 HDR CR 4.2, HDS-12 NDR CR 4.2

FRS row	HDS elements
FRS-CMP-403	HDS-12 EDR CR 4.3, HDS-12 HDR CR 4.3, HDS-12 NDR CR 4.3
FRS-CMP-501	HDS-12 EDR CR 5.1, HDS-12 HDR CR 5.1, HDS-12 NDR CR 5.1
FRS-CMP-601	HDS-12 EDR CR 6.1, HDS-12 HDR CR 6.1, HDS-12 NDR CR 6.1
FRS-CMP-602	HDS-12 EDR CR 6.2, HDS-12 HDR CR 6.2, HDS-12 NDR CR 6.2
FRS-CMP-701	HDS-12 EDR CR 7.1, HDS-12 HDR CR 7.1, HDS-12 NDR CR 7.1
FRS-CMP-702	HDS-12 EDR CR 7.2, HDS-12 HDR CR 7.2, HDS-12 NDR CR 7.2
FRS-CMP-703	HDS-12 EDR CR 7.3, HDS-12 HDR CR 7.3, HDS-12 NDR CR 7.3
FRS-CMP-704	HDS-12 EDR CR 7.4, HDS-12 HDR CR 7.4, HDS-12 NDR CR 7.4
FRS-CMP-706	HDS-12 EDR CR 7.6, HDS-12 HDR CR 7.6, HDS-12 NDR CR 7.6
FRS-CMP-707	HDS-12 EDR CR 7.7, HDS-12 HDR CR 7.7, HDS-12 NDR CR 7.7
FRS-CMP-708	HDS-12 EDR CR 7.8, HDS-12 HDR CR 7.8, HDS-12 NDR CR 7.8
FRS-DEV-E204	HDS-12 EDR 2.4
FRS-DEV-E213	CH-1…4, CP-n-DOOR, PH-02, PH-03, PH-04, HDS-10 ControlLogix 5580, HDS-10 Symbio 800, HDS-10 PowerFlex 755T / FC 102, HDS-12 EDR 2.13
FRS-DEV-E302	HDS-12 EDR 3.2
FRS-DEV-E310	HDS-12 EDR 3.10
FRS-DEV-E311	CP-n-DOOR, PH-01, PH-02, PH-03, HDS-12 EDR 3.11
FRS-DEV-E312	HDS-12 EDR 3.12
FRS-DEV-E313	HDS-12 EDR 3.13
FRS-DEV-E314	HDS-12 EDR 3.14
FRS-DEV-H204	HDS-12 HDR 2.4
FRS-DEV-H213	NW-06, HS-01, HS-03, HDS-12 HDR 2.13
FRS-DEV-H302	HDS-12 HDR 3.2
FRS-DEV-H310	HDS-12 HDR 3.10
FRS-DEV-H311	HS-04, HDS-12 HDR 3.11
FRS-DEV-H312	HS-01, HDS-10 Servers and workstations, HDS-12 HDR 3.12
FRS-DEV-H313	HDS-12 HDR 3.13
FRS-DEV-H314	ESX-5A, 5B, HS-01, HDS-10 Servers and workstations, HDS-10 Panel PC (OIS-1), HDS-12 HDR 3.14
FRS-DEV-N106	PH-06, HDS-12 NDR 1.6
FRS-DEV-N113	HDS-12 NDR 1.13
FRS-DEV-N204	HDS-12 NDR 2.4
FRS-DEV-N213	NW-03, PH-04, HDS-10 Stratix 5800, HDS-10 FortiGate, HDS-12 NDR 2.13
FRS-DEV-N302	HDS-12 NDR 3.2
FRS-DEV-N310	HDS-12 NDR 3.10
FRS-DEV-N311	HS-04, PH-02, HDS-12 NDR 3.11
FRS-DEV-N312	HDS-12 NDR 3.12
FRS-DEV-N313	HDS-12 NDR 3.13
FRS-DEV-N314	HDS-12 NDR 3.14
FRS-DEV-N502	HDS-12 NDR 5.2
FRS-DEV-N503	HDS-12 NDR 5.3
FRS-PLC-002	HDS-10 ControlLogix 5580

Annex B Open items to confirm

Carried from URS, FRS and SDS (OI-01 to OI-20) plus:

ID	Party	Item
OI-21	Rockwell	1756-L83E memory and connection sizing for the final I/O and OPC UA tag count; Stratix 5800 variant with DLR and PTP.
OI-22	Integrator	Final I/O counts, drive frames and motor sizes; panel heat load; UPS sizing.